

CW6 – QuPath for Multiplex Immunofluorescence Analysis

Pre-Workshop Installation Guide

SPAOM 2026 — Tuesday 6 October, 15:00. Wednesday 7 October, 12:20.

Consider that 10 laptops will be available for completing the CW, assigning each laptop to a pair of attendees. In contrast, if you prefer to take your own laptop, please complete this installation before the workshop. Software installation is not part of the session, so everything below should be done in advance, on the laptop you will bring to the workshop.

1. Before the workshop: Quick checklist

- Bring your own laptop (Windows, macOS, or Linux) with administrator rights to install software.
- Bring a computer mouse with a scroll wheel if possible. QuPath is much easier to use with a mouse than a trackpad; a limited number of spare mice will be available on site for those who don't have one.

- ***Minimum specification***

QuPath won't run properly with all features unless you have at least the following:

- 64-bit processor & operating system
- 4 GB RAM (for browsing images, limited analysis)
- 500 MB hard disk space
- For Mac, at least OS X 10.7.4

- ***Recommended specification***

Ideally, for better performance you would have the following:

- Multi-core processor (e.g. Intel i7)
 - 16 GB RAM
 - Discrete graphics card
 - Solid state hard disk
 - Lots of hard disk space to store images and data - may require up to 1 GB to store data related to a single slide
- Make sure your laptop is fully charged and bring your charger — power outlets on site are available.

- The full multiplex dataset ('*melanoma_raw.tiff*' as the original file (66.9 GB)), a lightweight test image, '*melanoma_crop.tiff*' as the cropped version (2.0 GB)), and this guide will be shared with you by email / cloud link prior to the workshop. Please download everything in advance.
 - **Note:** Depending on your computer specifications, the full **66.9 GB** dataset may **not be usable**. If your system does not meet the **minimum requirements**, please use the **2.0 GB** cropped dataset ('*melanoma_crop.tiff*'). For the **Community Workshop**, the **2.0 GB** dataset is strongly recommended for all participants.
- Complete all the steps in this guide (QuPath installation, Cellpose installation, and the final test in Section 4) before arriving. If you run into problems, contact us in advance so we can help (see contact details at the end of this document).

2. Install QuPath v0.7.0

We will use QuPath v0.7.0 (the current stable release). Please install this exact version so that everyone in the room has the same software — do not install an older or a pre-release build.

Downloads for all platforms are available from the official releases page: <https://github.com/qupath/qupath/releases> (look for the v0.7.0 release).

Windows

1. Download QuPath-v0.7.0-Windows.msi from the releases page above.
2. Double-click the .msi file and follow the default installer prompts.
3. Write QuPath in the Windows search bar and open the software to launch it and confirm it opens correctly.

macOS

1. Download the .pkg file that matches your Mac: QuPath-v0.7.0-Mac-arm64.pkg for Apple Silicon (M1/M2/M3/M4), or QuPath-v0.7.0-Mac-x64.pkg for an Intel Mac.
2. Double-click the .pkg file and follow the default installer prompts.
3. If macOS blocks the app ("cannot be opened because the developer cannot be verified"), go to System Settings → Privacy & Security and click "Open Anyway" next to the QuPath warning.
4. Launch QuPath from Applications to confirm it opens correctly.

Linux

1. Download QuPath-v0.7.0-Linux.tar.xz from the releases page above.
2. Extract the archive to a location of your choice, e.g. `tar -xf QuPath-v0.7.0-Linux.tar.xz`
3. Inside the extracted folder, make the launcher executable if needed (`chmod +x QuPath`) and run it from the `bin/` subfolder, or double-click it from your file manager.

Note: On some Ubuntu/Debian systems, QuPath extensions that call external programs (including the Cellpose extension in Section 3) can fail with a "Permission Denied (Error 13)" message. If you hit this, please get in touch before the workshop.

Note: The first time you open QuPath, it may ask whether you want to open the Extension Manager, and it will ask you to set a QuPath user directory. Click "Yes" to open the Extension Manager, and for the user directory you can simply keep the default location it suggests.

3. Install Cellpose and the QuPath Cellpose extension

Cellpose is not built into QuPath. It runs as a separate Python program, and QuPath talks to it through the “QuPath Cellpose” extension. This means there are two things to install: (1) Cellpose itself, in a Python environment, and (2) the QuPath extension that connects to it.

3.1 Install a Python environment with Cellpose

If you don't already have conda or Python installed, install Miniconda first: <https://docs.conda.io/en/latest/miniconda.html>.

Download Miniconda and follow the installer instructions.

Then open the anaconda prompt and run:

```
conda create -n cellpose-qupath python=3.10
conda activate cellpose-qupath
pip install cellpose
pip install scikit-image
```

Note: *scikit-image* is required by the extension's quality-control step. GPU acceleration is optional and not required for this workshop (CPU is fine for the dataset sizes we will use).

3.2 Get the path to your Python executable

With the environment still activated, run the command for your operating system and copy the resulting path

Windows

```
where python
```

macOS / Linux

```
which python
```

This will print something like C:\Users\yourname\miniconda3\envs\cellpose-qupath\python.exe (Windows) or /Users/yourname/miniconda3/envs/cellpose-qupath/bin/python (macOS/Linux). Save this path somewhere (you will need it in step 3.4.)

3.3 Install the QuPath Cellpose extension

1. Open QuPath, then go to Extensions → Manage Extensions.
2. Install and set up the QuPath BIOP catalog first, if prompted (this is what lets QuPath find and install the Cellpose extension).
 - On the Extension Manager window click Manage extension catalogs
 - Enter the catalog URL: <https://github.com/BIOP/qupath-biop-catalog>
 - Browse and install the extensions you need
3. From the QuPath BIOP catalog, find and install the “QuPath Cellpose v0.12.1” extension.
4. Restart QuPath after the installation completes.

Note: If your QuPath extensions directory is not yet set, QuPath will prompt you to choose a folder the first time you install an extension (Edit → Preferences → Extensions). Any empty folder works.

3.4 Connect QuPath to your Cellpose installation

1. In QuPath, go to Edit → Preferences → Cellpose/Omnipose.
2. Paste the Python executable path you copied in step 3.2 into the Cellpose environment field (“Cellpose ‘python.exe’ location”).
3. Leave the Omnipose field empty — we will not use Omnipose in this workshop.
4. Close the Preferences window.

4. Test that everything works

Please do this test before the workshop, not on the day. If it doesn't work, contact us in advance.

Step 1 — test Cellpose from the command line

With your environment activated, run:

```
python -m cellpose --version
```

This should print a version number without errors. If it does, Cellpose itself is correctly installed.

Step 2 — test the extension inside QuPath

1. Open the small test image we shared with you (see Section 1) in QuPath, and create a new project for it.
2. Draw or select an annotation on the image (e.g. a rectangle around a region containing cells).
3. Go to Extensions → Cellpose → Cellpose detection script template. This opens a ready-made script in the script editor.
4. Run the script (Run → Run, or the Run button) with the default settings.
5. If Cellpose is correctly installed and connected, you should see cell detections appear inside your annotation after a short delay (the first run may take longer, as Cellpose downloads its pretrained model).

Note: If nothing happens the first time, try running Cellpose once from the command line with the default model (e.g. `python -m cellpose --pretrained_model cyto2`) so the pretrained model downloads outside QuPath, then re-run the script in QuPath.

If both steps above work, you are fully set up for the workshop — no further installation will be needed on the day.

5. Questions or problems before the workshop?

If you get stuck at any point, please contact us before 6 October so we can help you troubleshoot in advance as there won't be time to solve installation problems from scratch during the session:

- María Calvo de Mora Román — mcalvo@cnio.es
- Ana Cayuela — acayuela@cnio.es

See you at CW6!